Functions & Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A car travels 120 miles in 3 hours. Find the average speed.

Apply the fundamental distance-rate-time relationship $d = r \cdot t$: $r = \frac{d}{t} = \frac{120 \text{ miles}}{3 \text{ hours}} = \mathbf{40}$ mph ACT Exam Note: Average speed is always Total Distance divided by Total Time. Never average two speeds directly if the travel times are different.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple of 6 and 8.

Prime Factorization Method: $6 = 2^1 \times 3^1$ $8 = 2^3$ The Least Common Multiple (LCM) takes the highest power of every prime factor present: $\text{LCM} = 2^3 \times 3^1 = 8 \times 3 = \mathbf{24}$ ACT Tip: List multiples of the larger number until divisible by the smaller: $8, 16, 24$ (divisible by 6).

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Give the equation of the horizontal line through $(0, 5)$.

A horizontal line has a slope of $m = 0$ and every point on it shares the same y-coordinate. Therefore, its general equation is $y = c$. Since it passes through $(0, 5)$, the equation is $\mathbf{y = 5}$. Contrast: A vertical line through the same point would have undefined slope and equation $x = 0$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the distance between $(0, 0)$ and $(3, 4)$.

Apply the Euclidean distance formula derived from the Pythagorean theorem: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = \mathbf{5}$ ACT Shortcut: Recognize the fundamental $3 - 4 - 5$ Pythagorean triple immediately.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the interior-angle sum of a pentagon.

Use the polygon interior angle sum formula for an n-gon: $S = (n - 2) \times 180^\circ$ A pentagon has $n = 5$ sides: $S = (5 - 2) \times 180^\circ = 3 \times 180^\circ = \mathbf{540}^\circ$ Bonus: If regular, each interior angle measures $\frac{540^\circ}{5} = 108^\circ$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For what value of x is $\frac{1}{x-2}$ undefined?

A rational expression is undefined wherever its denominator equals zero (division by zero is undefined in real numbers). Set the denominator to zero: $x - 2 = 0 \Rightarrow \mathbf{x = 2}$ At $x = 2$, the graph of $y = \frac{1}{x-2}$ has a vertical asymptote.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(2))$.

Evaluate composite functions working from the inside out: Step 1: Evaluate the inner function $g(2)$: $g(2) = 2^2 = 4$ Step 2: Substitute this result as the input into the outer function $f(x)$: $f(g(2)) = f(4) = 2(4) + 1 = 8 + 1 = \mathbf{9}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 6 and adjacent 8. Find $\tan\theta$.

Recall the tangent ratio from SOH CAH TOA: $\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{6}{8}$ Reduce the fraction by dividing numerator and denominator by 2: $\tan\theta = \frac{3}{4} = \mathbf{0.75}$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the midpoint of $(2, 4)$ and $(6, 8)$.

Apply the midpoint formula by averaging the coordinates: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{2+6}{2}, \frac{4+8}{2}\right) = \left(\frac{8}{2}, \frac{12}{2}\right) = \mathbf{(4, 6)}$

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the circumference of a circle with $r = 7$. Take $\pi = \frac{22}{7}$.

Apply the circle circumference formula $C = 2\pi r$: $C = 2 \times \frac{22}{7} \times 7$ Cancel the factor of 7 in the numerator and denominator: $C = 2 \times 22 = \mathbf{44}$

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11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of ΔABC? Show your complete algebraic work.

Final Answer: 15. To find the area of ΔABC, we first determine the coordinates of its three vertices: A, B, and C. 1. **Vertex C**: The function $f(x) = 9 - x^2$ is a downward-opening parabola with its maximum at $x = 0$. $f(0) = 9 - 0^2 = 9$, so the vertex is $C(0, 9)$. 2. **Intersection Points A and B**: Set $f(x) = g(x)$: $9 - x^2 = x + 3$ Rearrange into standard quadratic form: $x^2 + x - 6 = 0$ Factor the quadratic equation: $(x + 3)(x - 2) = 0$ Thus, the x-coordinates of the intersection points are $x = -3$ and $x = 2$. - For $x = -3$: $y = g(-3) = -3 + 3 = 0$, giving point $A(-3, 0)$. - For $x = 2$: $y = g(2) = 2 + 3 = 5$, giving point $B(2, 5)$. 3. **Area of ΔABC**: Using the coordinate geometry area formula (Shoelace formula) for vertices $(-3, 0)$, $(2, 5)$, and $(0, 9)$: Area $= \frac{1}{2}|x_A(y_B - y_C) + x_B(y_C - y_A) + x_C(y_A - y_B)|$ Substitute the coordinates: Area $= \frac{1}{2}|-3(5 - 9) + 2(9 - 0) + 0(0 - 5)|$
Area $= \frac{1}{2}|-3(-4) + 2(9) + 0| = \frac{1}{2}|12 + 18| = \frac{1}{2}(30) = 15$ Alternatively, enclosing ΔABC in a bounding rectangle with horizontal span from $x = -3$ to $x = 2$ (width 5) and vertical span from $y = 0$ to $y = 9$ (height 9) gives a total box area of $5 \times 9 = 45$. Subtracting the three surrounding right triangles gives $45 - \frac{1}{2}(5)(5) - \frac{1}{2}(2)(4) - \frac{1}{2}(3)(9) = 45 - 12.5 - 4 - 13.5 = 15$. Common errors include forgetting the factor of $\frac{1}{2}$ (yielding 30) or multiplying the horizontal width by the total height and dividing by 2 without adjusting for the non-right shape (yielding 22.5).

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(1, -3)$ and passing through the point $(3, 1)$. What is the value of $f(-2)$? Show your complete algebraic work.

Final Answer: 6. Step 1: Identify the parameters from the vertex. The vertex of the parabola is given as $(h, k) = (1, -3)$. Substituting these values into the vertex form gives: $f(x) = a(x - 1)^2 - 3$ Step 2: Solve for the leading coefficient a. Since the graph passes through $(3, 1)$, substitute $x = 3$ and $f(3) = 1$: $1 = a(3 - 1)^2 - 3$ $1 = a(2)^2 - 3$ $1 = 4a - 3$ $4a = 4 \Rightarrow a = 1$ Thus, the function is $f(x) = (x - 1)^2 - 3$. Step 3: Evaluate $f(-2)$. Substitute $x = -2$ into the equation:
$f(-2) = (-2 - 1)^2 - 3 = (-3)^2 - 3 = 9 - 3 = 6$ Common Misconceptions: - Forgetting to square the difference when solving for a (treating $(3 - 1)^2$ as 2) leads to incorrect scaling. - Neglecting the vertical shift -3 leads to $(-3)^2 = 9$. - Confusing the signs in the vertex form $(x - h)$ as $(x + 1)$ leads to $(-2 + 1)^2 - 3 = -2$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(3, 2)$ and passing through the point $(1, 6)$. What is the value of b? Show your complete algebraic work.

Final Answer: -6. 1. Write the quadratic function in vertex form: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. 2. Given the vertex $(3, 2)$, the equation becomes: $f(x) = a(x - 3)^2 + 2$ 3. Use the given point $(1, 6)$ to solve for the parameter a: $6 = a(1 - 3)^2 + 2$ $6 = a(-2)^2 + 2$ $6 = 4a + 2 \Rightarrow 4a = 4 \Rightarrow a = 1$ 4. Substitute $a = 1$ into the vertex form and expand to standard form $f(x) = ax^2 + bx + c$: $f(x) = 1(x - 3)^2 + 2 = (x^2 - 6x + 9) + 2 = x^2 - 6x + 11$ 5. Comparing coefficients with $ax^2 + bx + c$, we identify $a = 1$, $b = -6$, and $c = 11$. Therefore, $b = -6$. Common traps: - Incorrectly assuming $b = -h = -3$ or $b = h = 3$ directly from the vertex coordinates without expanding into standard form. - Making a sign error when expanding $(x - 3)^2$ as $x^2 + 6x + 9$, which leads to $b = 6$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c? Show your complete algebraic work.

Final Answer: 4. 1. Identify the vertex form of the parabola: Since the vertex is $(h, k) = (2, -4)$, the function is: $f(x) = a(x - 2)^2 - 4$ 2. Solve for the leading parameter a using the given y-intercept $(0, -2)$: $-2 = a(0 - 2)^2 - 4$ $-2 = 4a - 4$ $2 = 4a \Rightarrow a = \frac{1}{2}$ 3. Determine the full equation of the function: $f(x) = \frac{1}{2}(x - 2)^2 - 4$ 4. Find c by substituting $x = 6$: $c = f(6) = \frac{1}{2}(6 - 2)^2 - 4 = \frac{1}{2}(16) - 4 = 8 - 4 = 4$ Misconception traps: - Assuming $a = 1$ leads to $(6 - 2)^2 - 4 = 12$ (Choice D). - Forgetting to subtract $k = 4$ leads to $\frac{1}{2}(16) = 8$ (Choice C).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 2}$? Show your complete algebraic work.

Need $x - 2 \geq 0 \Rightarrow x \geq 2$. Smallest value: 2.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

The graph of $y = -f(x)$ is the graph of $y = f(x)$ transformed how? Show your complete algebraic work.

Final Answer: reflected across the x-axis. Negating the output $f(x)$ flips the graph vertically: reflection across the x-axis.

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

The graph of $y = f(-x)$ is the graph of $y = f(x)$ transformed how? Show your complete algebraic work.

Final Answer: reflected across the y-axis. Negating the input x flips the graph horizontally: reflection across the y-axis.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

The graph of $y = f(x - 5)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

Final Answer: 5 units right. $y = f(x - h)$ shifts right when $h > 0$, left when $h < 0$; here shifts 5 units right.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

The graph of $y = f(x - 2)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

Final Answer: 2 units right. $y = f(x - h)$ shifts right when $h > 0$, left when $h < 0$; here shifts 2 units right.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

The graph of $y = f(x - 1)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

Final Answer: 1 units right. $y = f(x - h)$ shifts right when $h > 0$, left when $h < 0$; here shifts 1 units right.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $f(x) = 3x^2 - 4x + 7$, what is the value of $f(-2)$?

a. 11

b. 21

c. 27 ✓

d. 31

$f(-2) = 3(-2)^2 - 4(-2) + 7 = 3(4) + 8 + 7 = 12 + 8 + 7 = 27$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the linear function $f(x) = \frac{4x - 7}{3}$, what is the equation of the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x + 7}{4}$ ✓

b. $f^{-1}(x) = \frac{3x - 7}{4}$

c. $f^{-1}(x) = \frac{4x + 3}{7}$

d. $f^{-1}(x) = \frac{7x + 3}{4}$

Set $y = \frac{4x - 7}{3} \Rightarrow 3y = 4x - 7 \Rightarrow 4x = 3y + 7 \Rightarrow x = \frac{3y + 7}{4}$. Swapping variables gives $f^{-1}(x) = \frac{3x + 7}{4}$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $g(x) = \sqrt{2x + 10}$, what is the domain of $g(x)$ in the set of real numbers?

a. $x \geq -5$ ✓

b. $x > -5$

c. $x \geq 0$

d. $x \leq -5$

The expression under the radical must be non-negative: $2x + 10 \geq 0 \Rightarrow 2x \geq -10 \Rightarrow x \geq -5$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$. What is the value of $(g \circ f)(1)$?

a. 11

b. 46 ✓

c. 49

d. 52

Evaluate inner function first: $f(1) = 2(1) + 5 = 7$. Then $(g \circ f)(1) = g(7) = 7^2 - 3 = 49 - 3 = 46$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $h(x) = 3x - 1$ and $(h \circ k)(x) = 6x + 8$, which of the following defines $k(x)$?

a. $k(x) = 2x + 3$ ✓

b. $k(x) = 2x + 9$

c. $k(x) = 3x + 4$

d. $k(x) = 2x + 2$

$h(k(x)) = 3k(x) - 1 = 6x + 8 \Rightarrow 3k(x) = 6x + 9 \Rightarrow k(x) = 2x + 3$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the range of the quadratic function $f(x) = -(x - 4)^2 + 9$?

a. $y \leq 9$ ✓

b. $y \geq 9$

c. $y \leq 4$

d. All real numbers

The parabola opens downward because of the negative leading coefficient. The vertex is at $(4, 9)$, which is a maximum. Thus the range is $y \leq 9$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = f(x)$ is shifted 3 units to the right and 5 units down. What is the equation of the transformed graph?

a. $y = f(x - 3) - 5$ ✓

b. $y = f(x + 3) - 5$

c. $y = f(x - 3) + 5$

d. $y = f(x + 3) + 5$

A horizontal shift to the right by h replaces x with $x - h$. A downward shift of k subtracts k from the function. Thus $y = f(x - 3) - 5$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following functions is an even function, satisfying $f(-x) = f(x)$ for all real x?

a. $f(x) = x^3 - x$

b. $f(x) = 4x^2 + 7$ ✓

c. $f(x) = 2x + 1$

d. $f(x) = x^4 - 3x$

Check $f(x) = 4x^2 + 7$: $f(-x) = 4(-x)^2 + 7 = 4x^2 + 7 = f(x)$. Therefore, it is even.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the horizontal asymptote of the rational function $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 3x - 4}$?

a. $y = 0$

b. $y = 3$ ✓

c. $y = 2$

d. No horizontal asymptote

Since degrees of numerator and denominator are equal (both 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{6}{2} = 3$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function is defined piecewise by $f(x) = 2x + 3$ for $x < 1$, and $f(x) = x^2 - 1$ for $x \geq 1$. What is the value of $f(-2) + f(3)$?

a. 7 ✓

b. 8

c. 9

d. 11

For $x = -2 < 1$: $f(-2) = 2(-2) + 3 = -1$. For $x = 3 \geq 1$: $f(3) = 3^2 - 1 = 8$. Sum $= -1 + 8 = 7$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $[x]$ denotes the greatest integer function (floor function), what is the value of $[3.8] - [-2.4]$?

a. 5

b. 6 ✓

c. 7

d. 1

$[3.8] = 3$ and $[-2.4] = -3$ (the greatest integer less than or equal to -2.4). Therefore, $3 - (-3) = 6$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c is the piecewise function $f(x) = cx + 4$ for $x \leq 2$, and $f(x) = 3x^2 - 2$ for $x > 2$ continuous at $x = 2$?

a. 2

b. 3 ✓

c. 4

d. 5

For continuity at $x = 2$, the left limit must equal the right limit:
$c(2) + 4 = 3(2)^2 - 2 \Rightarrow 2c + 4 = 12 - 2 = 10 \Rightarrow 2c = 6 \Rightarrow c = 3$.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = 4f(x)$ is the graph of $y = f(x)$ transformed how?

a. shifted up 4 units

b. **vertically stretched by a factor of 4** ✓

c. vertically compressed by a factor of 4

d. horizontally stretched by a factor of 4

Multiplying the output by 4 stretches the graph vertically by a factor of 4.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) - 2$ is the graph of $y = f(x)$ shifted how?

a. 5 units down

b. 1 units up

c. 2 units up

d. **2 units down** ✓

$y = f(x) + k$ shifts up when $k > 0$, down when $k < 0$; here shifts 2 units down.

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) + 8$ is the graph of $y = f(x)$ shifted how?

a. 8 units up ✓

b. 8 units down

c. 11 units up

d. 7 units down

$y = f(x) + k$ shifts up when $k > 0$, down when $k < 0$; here shifts 8 units up.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) + 6$ is the graph of $y = f(x)$ shifted how?

a. 5 units down

b. 9 units up

c. 6 units down

d. 6 units up ✓

$y = f(x) + k$ shifts up when $k > 0$, down when $k < 0$; here shifts 6 units up.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) + 2$ is the graph of $y = f(x)$ shifted how?

a. 2 units down

b. 1 units down

c. 5 units up

d. 2 units up ✓

$y = f(x) + k$ shifts up when $k > 0$, down when $k < 0$; here shifts 2 units up.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

The graph of $y = f(x) + 1$ is the graph of $y = f(x)$ shifted how?

a. 1 units down

b. 4 units up

c. 1 units up ✓

d. 5 units down

$y = f(x) + k$ shifts up when $k > 0$, down when $k < 0$; here shifts 1 units up.

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

What is the domain restriction for $f(x) = \frac{x+1}{x-9}$? (Give the excluded x-value.)

a. 11

b. 14

c. 3

d. **9 ✓**

The denominator $x - 9$ is zero when $x = 9$; this value is excluded.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 11}$?

a. **11 ✓**

b. 10

c. 26

d. 13

Need $x - 11 \geq 0 \Rightarrow x \geq 11$. Smallest value: 11.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 24}$?

a. 43

b. 23

c. 7

d. **24 ✓**

Need $x - 24 \geq 0 \Rightarrow x \geq 24$. Smallest value: 24.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 26}$?

a. **26 ✓**

b. 27

c. 12

d. 30

Need $x - 26 \geq 0 \Rightarrow x \geq 26$. Smallest value: 26.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 28}$?

a. 52

b. 58

c. 7

d. **28** ✓

Need $x - 28 \geq 0 \Rightarrow x \geq 28$. Smallest value: 28.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 8}$?

a. 34

b. 14

c. 16

d. **8** ✓

Need $x - 8 \geq 0 \Rightarrow x \geq 8$. Smallest value: 8.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

What value of x is excluded from the domain of $f(x) = \frac{1}{x}$?

a. -1

b. 1

c. 5

d. **0** ✓

The denominator is zero when $x = 0$, so $x = 0$ is excluded.

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